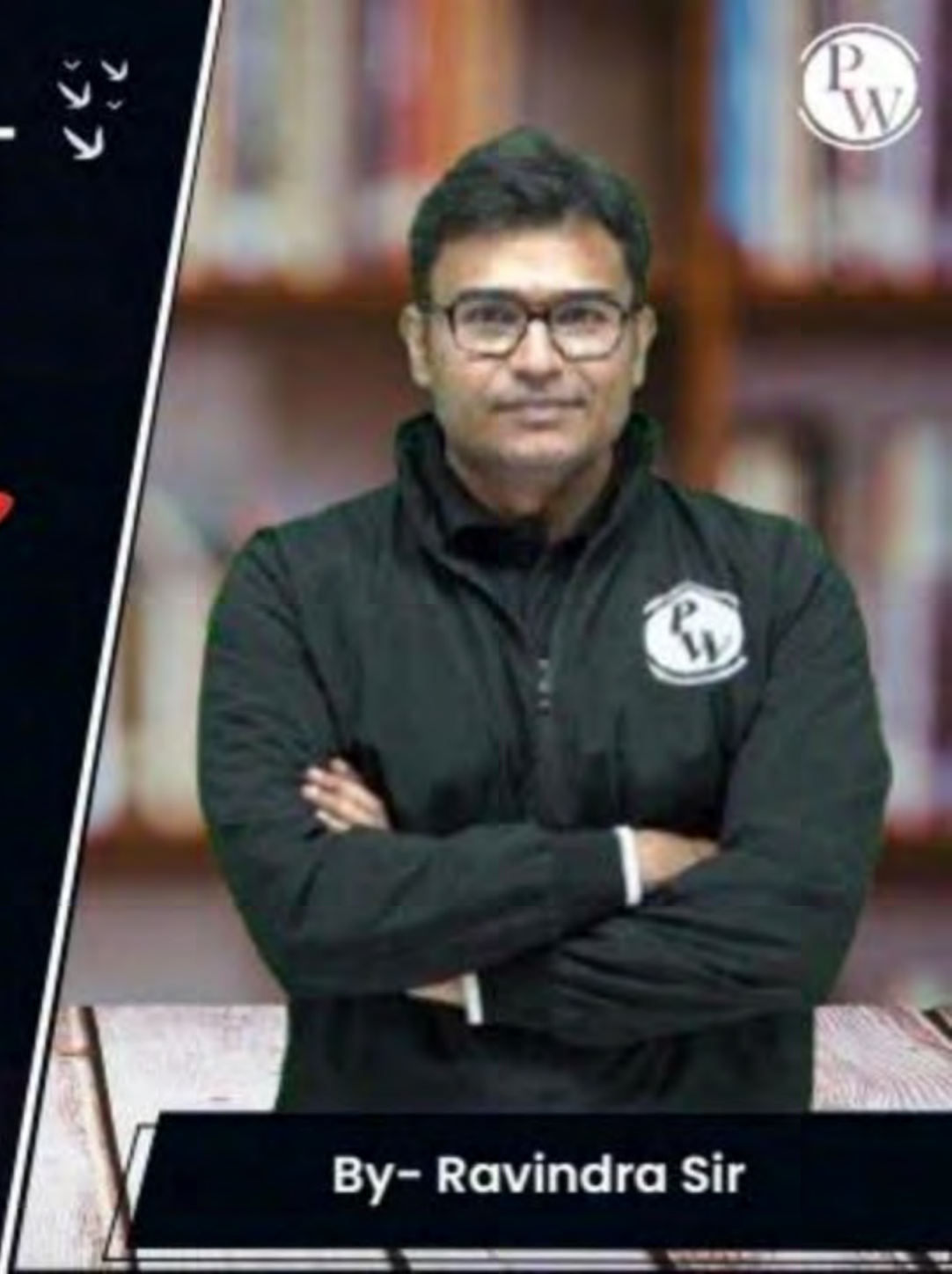
Computer Science & IT

Compiler Design



Introduction

Lecture No. 02



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

S
A
B
C
D
F

RBR

III - 86

IV - 13

IISc ME

ME thesis - Compilers

C - compiler - code optimization

S-grade

Topics to be Covered



- Topic
- Topic
- Topic

CD - 11Sc - YNS

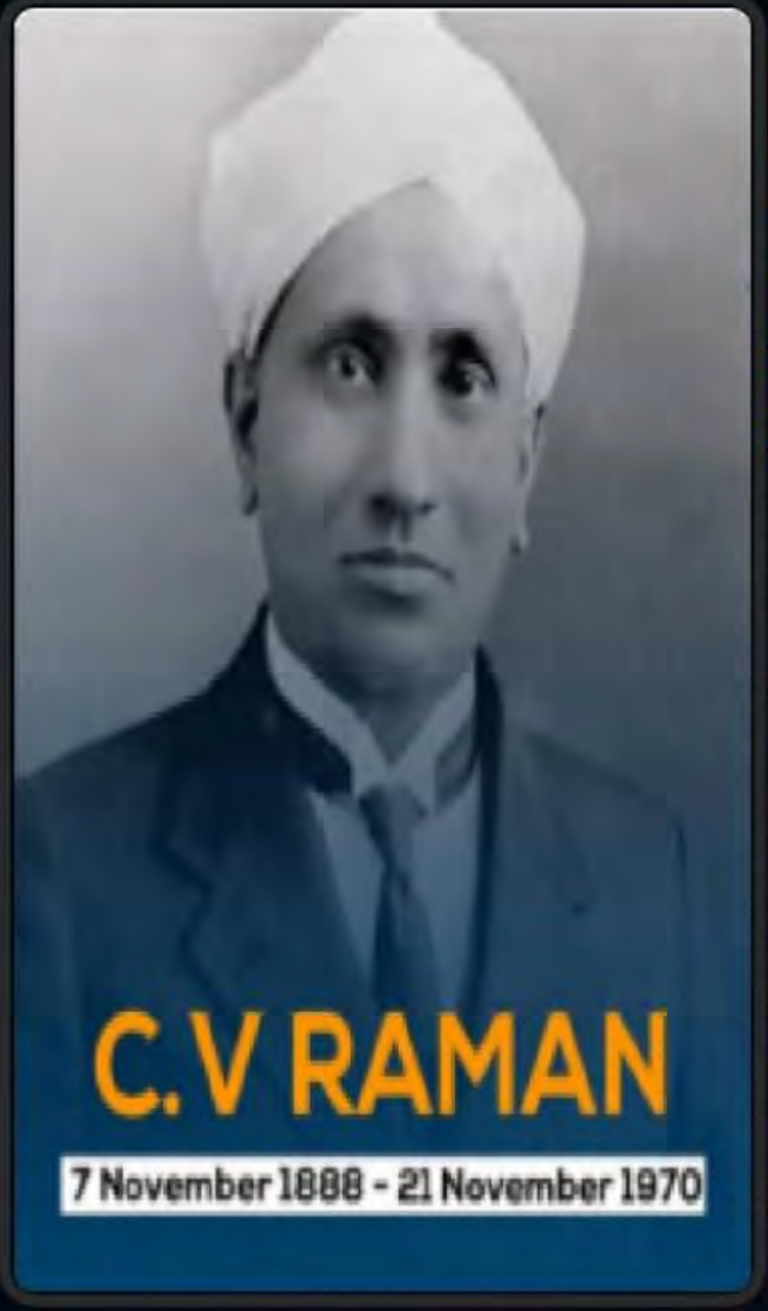
Assignment → Design (Compl

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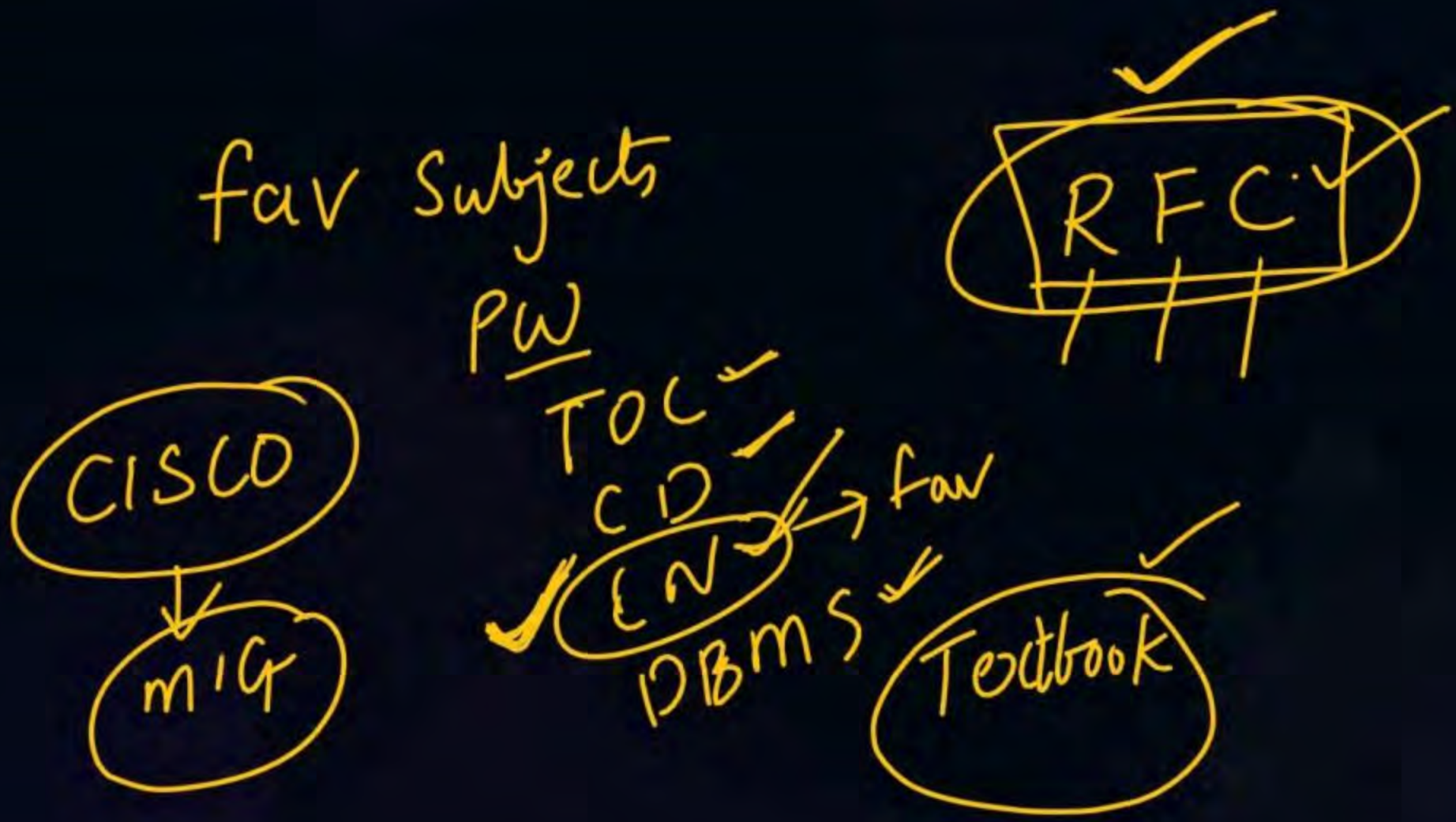
3 months Uney

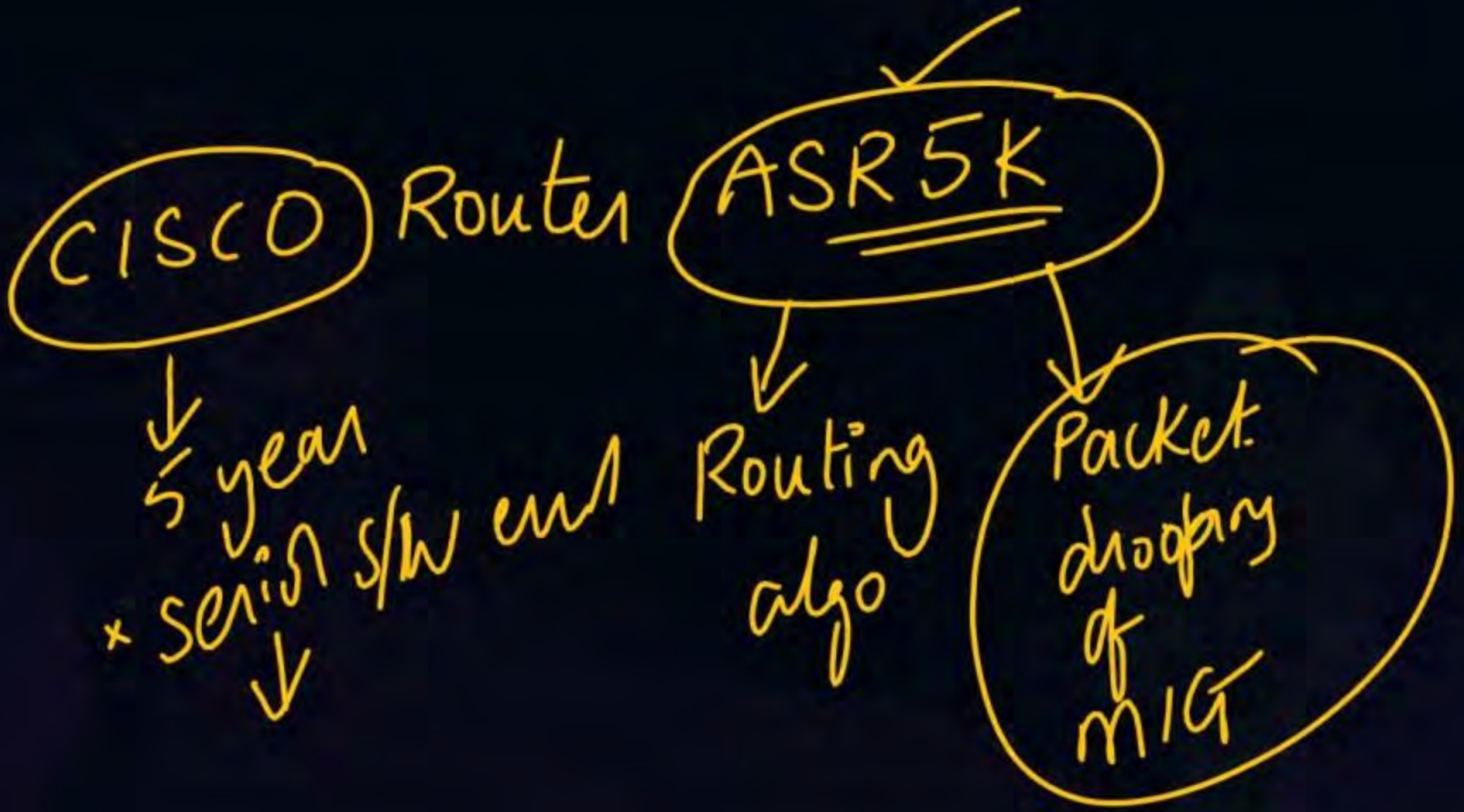
90% CN ✓



Unstoppable Indians: Stories to Ignite Student Motivation

C. V. Raman – The Raman Effect & Spectrometer Pioneer C. V. Raman (1888–1970) discovered the Raman effect in 1928, showing how light scatters in molecules and shifts in wavelength. Working at the Indian Association for the Cultivation of Science in Calcutta, he built a custom spectrometer—modifying a telescope, mirrors, and diffraction grating—to capture these subtle shifts. His painstaking experiments on crushed crystals confirmed inelastic light scattering. Awarded the 1930 Nobel Prize in Physics, Raman's work placed India on the global scientific map and inspired generations to innovate with limited resources.

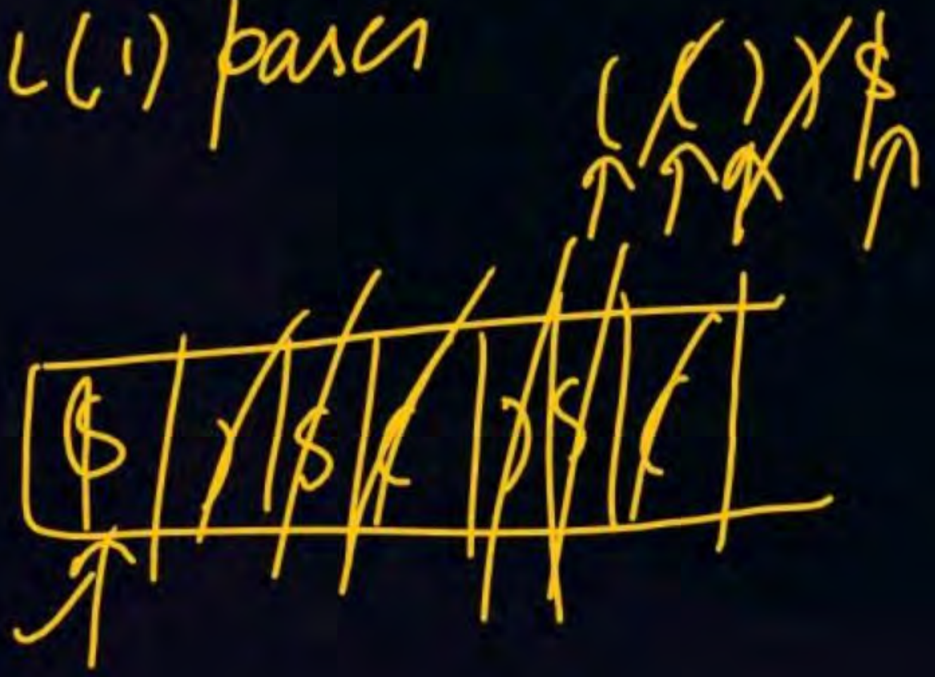

fav Subjects
PW
TOC
CD
CN
fav
DBMS
RFC
CISCO
MIG
Textbook


CISCO Router
ASR5K
5 year
× serious s/w error
Routing algo
Packet dropping of MIG

$S \rightarrow (S)/\epsilon$

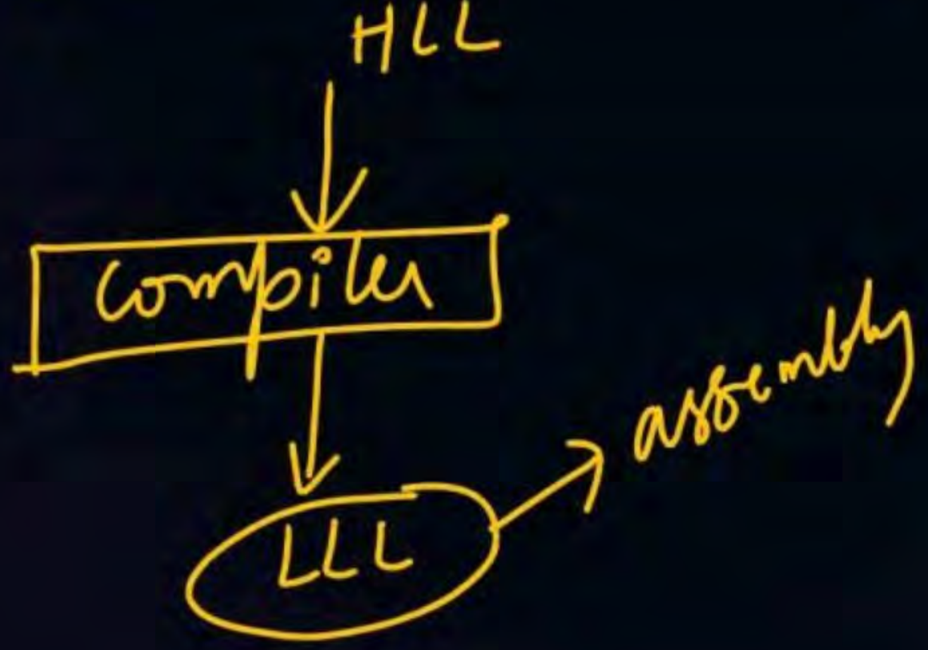
LL(1) parser

algo



Compiler design

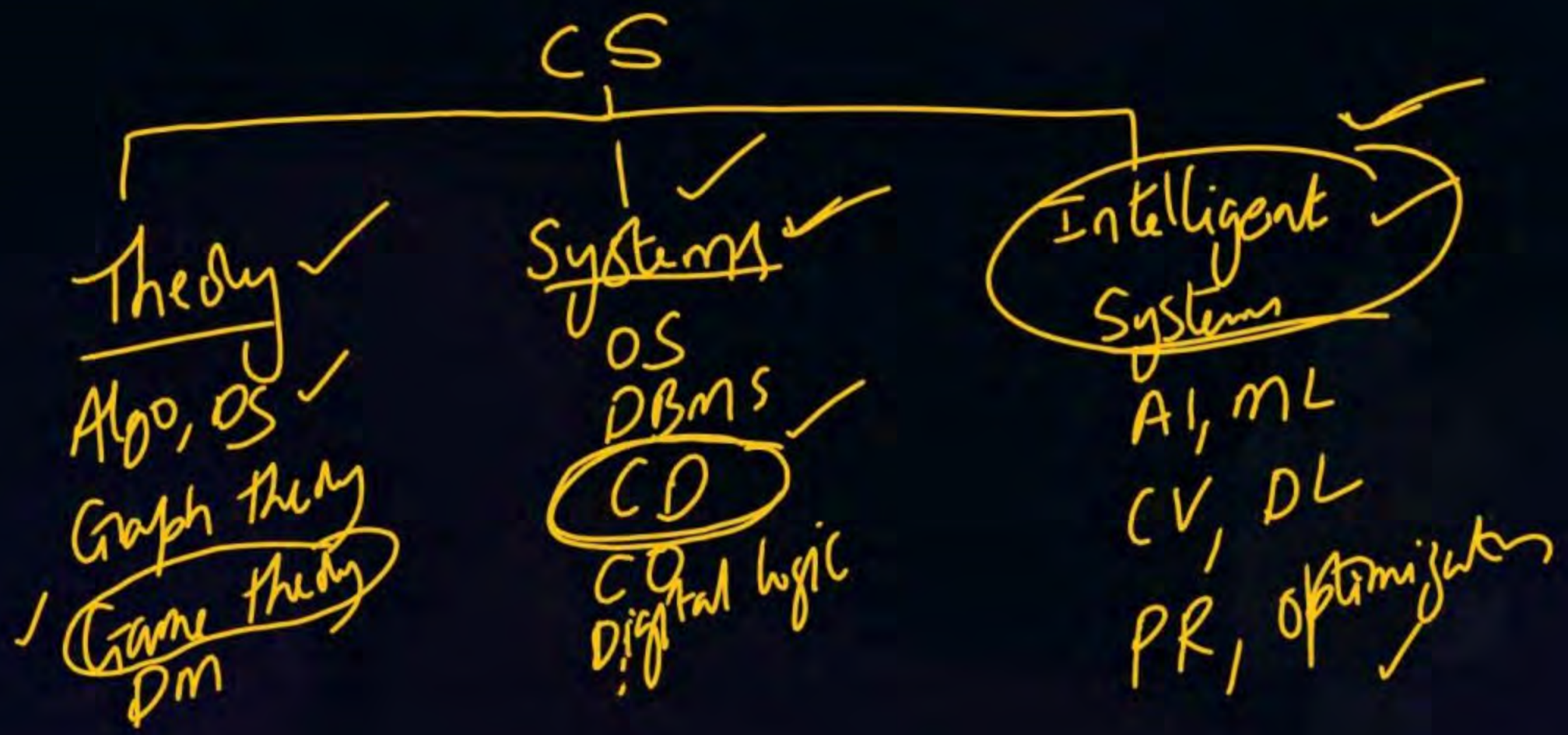
The main aim of compiler is to convert high level language into low level language



→ John Backus at IBM took 18 years to build first complete compiler for Fortran in 1957.

→ Students at IISc, builds C-compiler in 4 months

Building a compiler requires knowledge of



Building a compiler requires knowledge of

→ Programming languages (Parameter passing, Scope, memory allocation etc)

→ Theory (automata, context free languages)

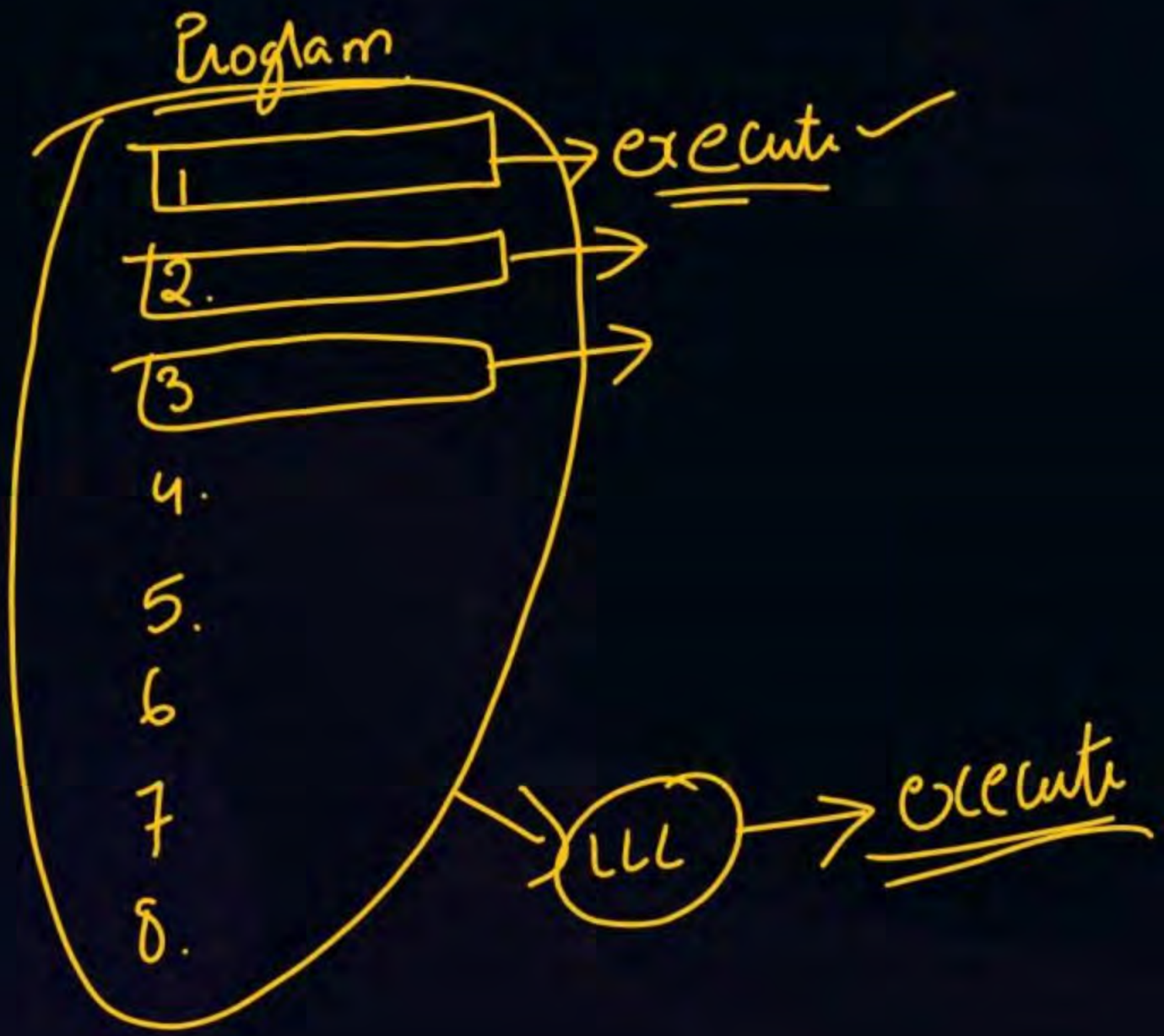
→ Algorithm and data structures (hash tables, graph algo, dynamic programming)

→ Computer architecture (assembly langues)

→ Software engineering.

Interpreter :-

An interpreter works similar to a compiler in most of the phases like lexical, syntax and semantic analyzers this analysis is performed for each single statement and when the statement is error free, instead of generating the corresponding code, the statement is executed and the result of execution is displayed.


Program
1
2.
3
4.
5.
6
7
8.
LLL
execute
execute

In interpreter every time a program has to be executed, it is processed and checked for errors, it is slower than compiled program

-> Some languages like Java are both Compiled and interpreted
The java program is first Compiled into byte code.
This byte code is interpreted



Unstoppable Indians: Stories to Ignite Student Motivation

Arogyaswami Paulraj – MIMO Wireless Innovator

Professor Arogyaswami Paulraj (b. 1944) invented MIMO (Multiple Input, Multiple Output) technology in 1992 at the Indian Institute of Science, Bangalore. He developed algorithms enabling multiple antennas at both transmitter and receiver ends to dramatically boost data rates and reliability. Through IIT Madras's technology transfer office, his patents paved the way for LTE and modern Wi Fi standards worldwide. Today's high speed mobile broadband networks rely on his work, and he has received numerous IEEE and governmental honors for revolutionizing wireless communications.

Adv of interpreter :-

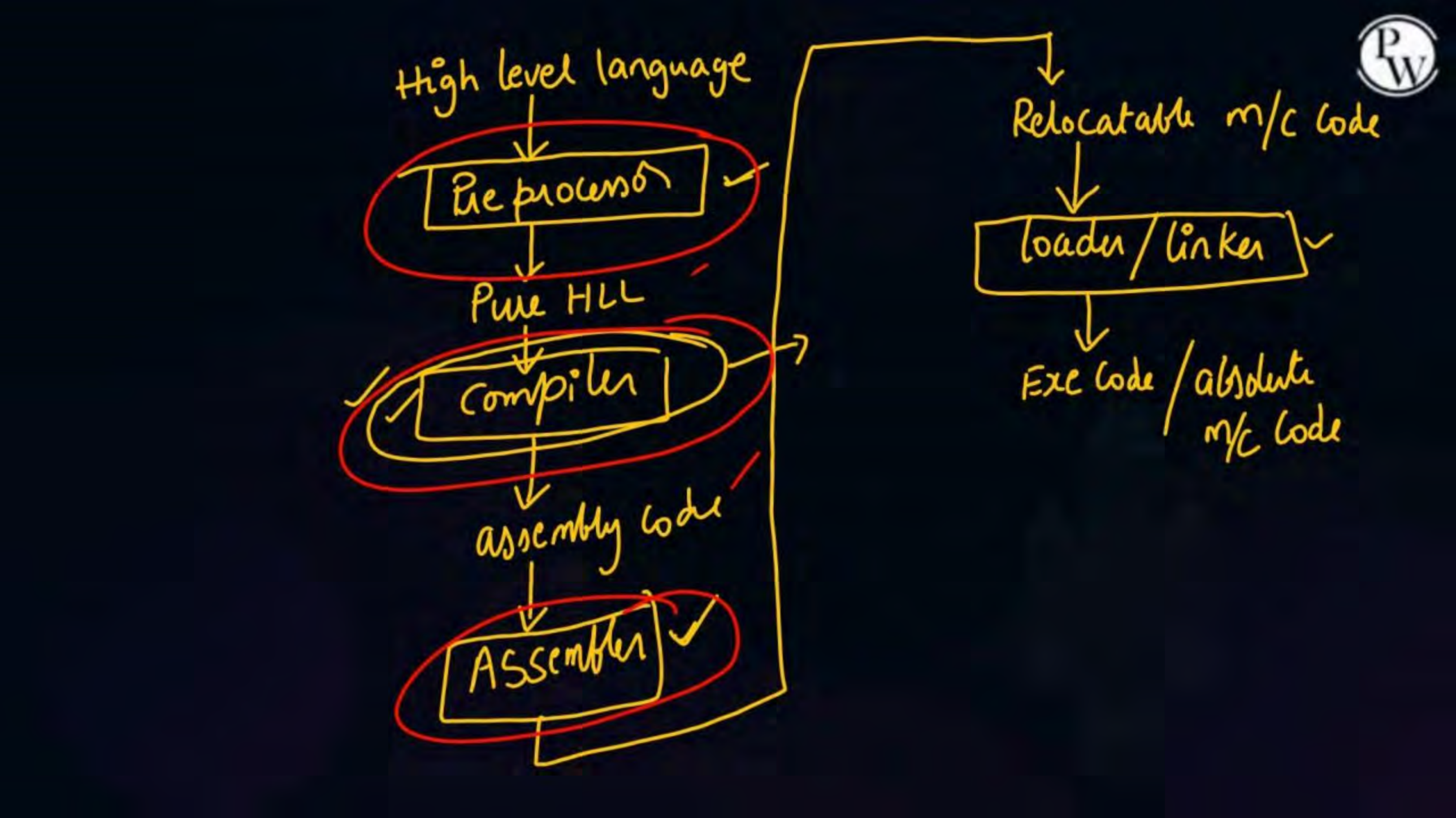
-> If a program is simple and is to be executed only once, an interpreter is preferred

-> Interpreters are more portable than compilers.



Certain language features are supported by interpretors

Ex:- dynamic types. In SNOBOL, the type of a variable is dependent on the value that is assigned to it any time


High level language
Pre processor
Pure HLL
Compiler
assembly code
Assembler
Relocatable m/c code
loader / linker
Exe code / absolute m/c code

Preprocessor :-

a) macro processing :- It performs macro expansion. All macros are identified and substituted with their respective values

Ex :- #define PI 3.14

b) File inclusion :- The statements like #include <studio.h> will be replaced by the files

c) Language extensions :- A preprocessor can add new functionalities to extend the language. Ex :- adding database quering facilities to the language

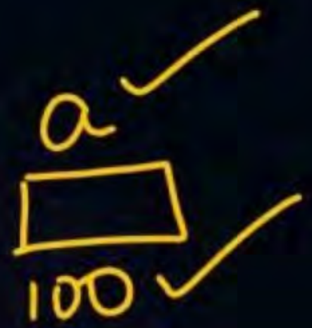

#define
#include
Pure HLL

Assembler :-

Assembly code is mnemonic version of m/c code in which names rather than the binary values for m/c instructions and memory addresses are used

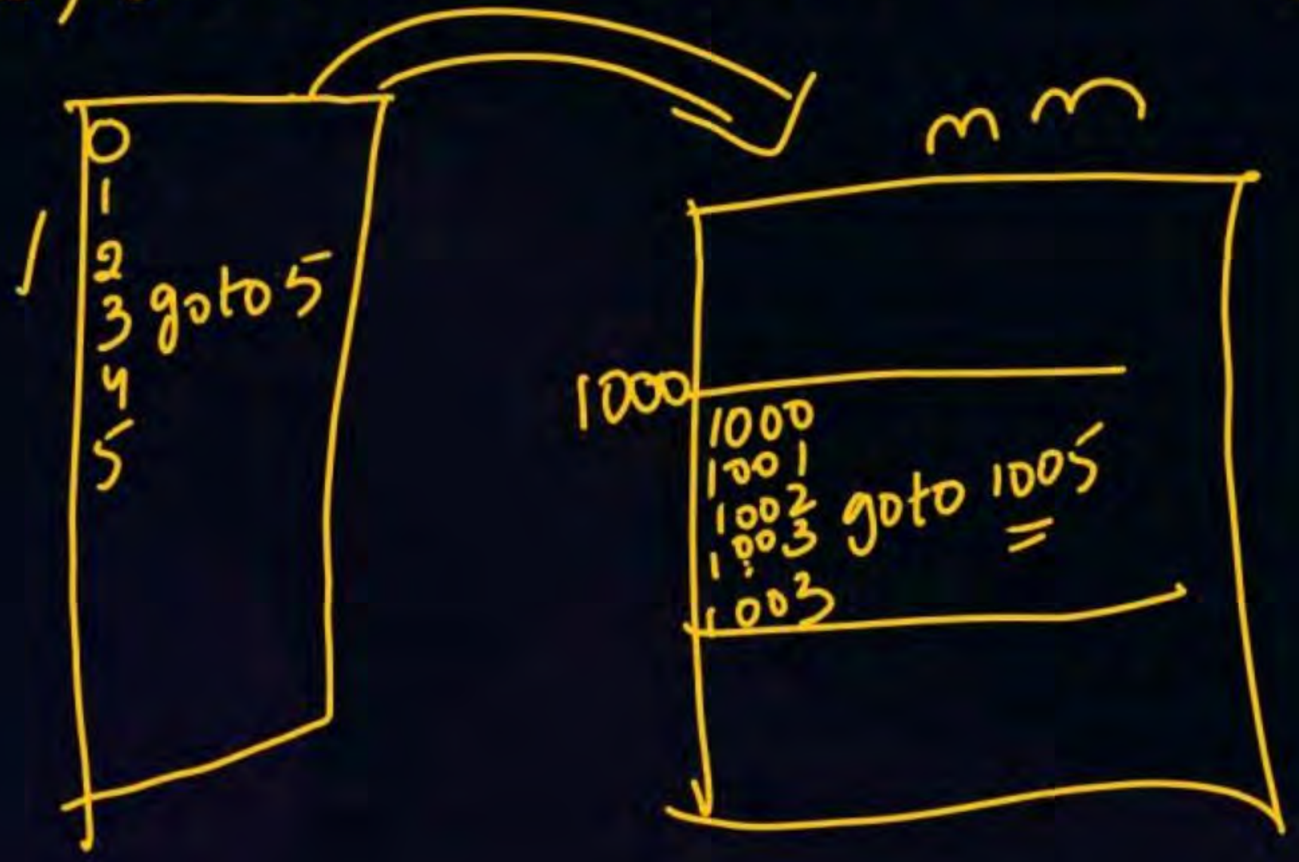


An assembler need to assign memory locations or addresses to the the symbols/identifiers.

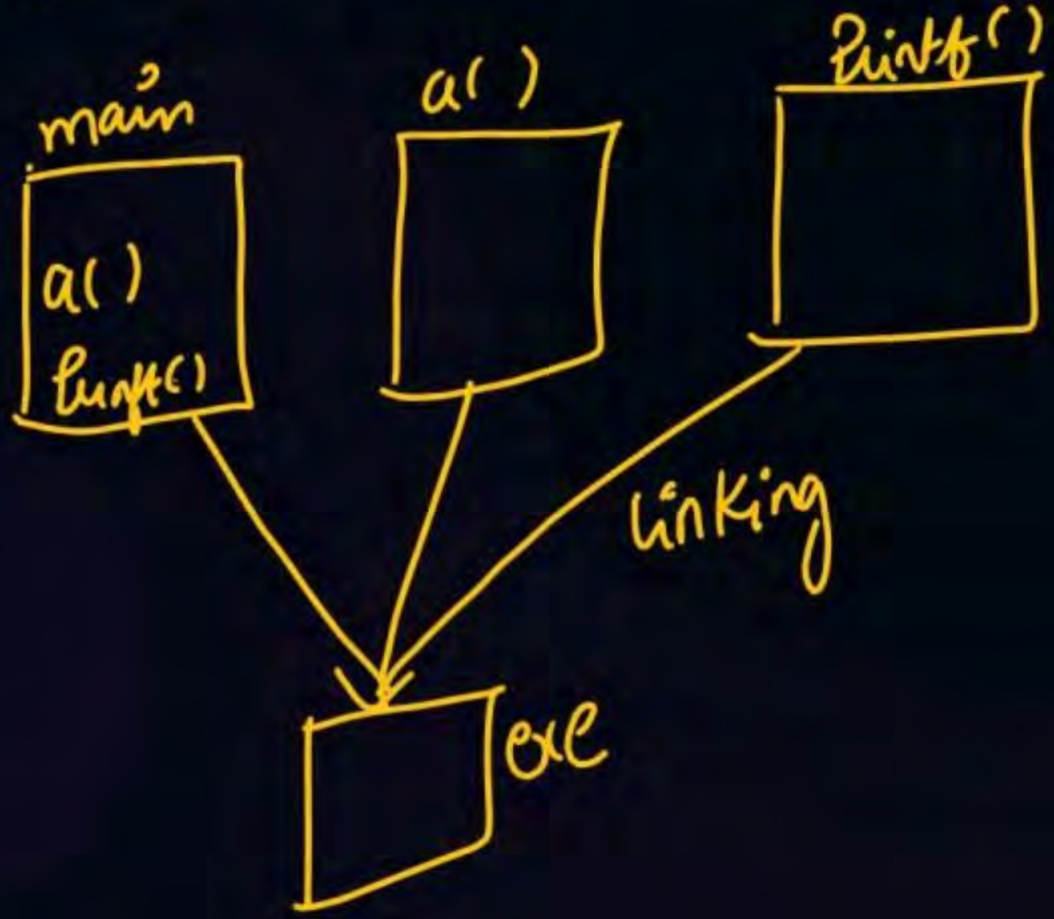


Loader / Linker :-

To convert the relocatable m/c code to executable Code, one of mode translator are required

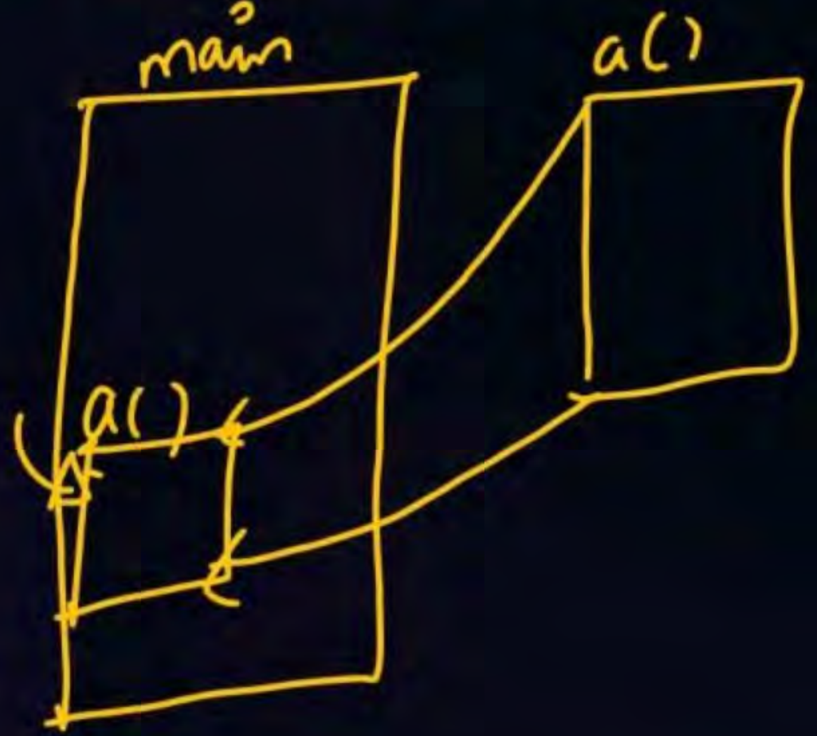


linking:- This is a process where a linker takes several object files and libraries as I/p and produces one file as output



Linker :-

It retrieves from the I/p files the code of all procedures that are referenced and resolves all external references to actual machine addresses





Unstoppable Indians: Stories to Ignite Student Motivation

P. K. Sethi – Jaipur Foot Inventor **Dr. P. K. Sethi (1927–2008), an orthopedic surgeon at SMS Medical College, Jaipur, co invented the Jaipur Foot in 1969. Crafted from locally sourced rubber and wood, this low cost, flexible prosthesis replicates natural movement—allowing users to walk, squat, and run. Partnering with craftsman Ram Chander Sharma, Sethi developed a modular assembly method that rural artisans could produce. Distributed via a cost recovery model, over 1.5 million Jaipur Feet have been fitted globally, showcasing how frugal engineering and social innovation can transform millions of lives.**

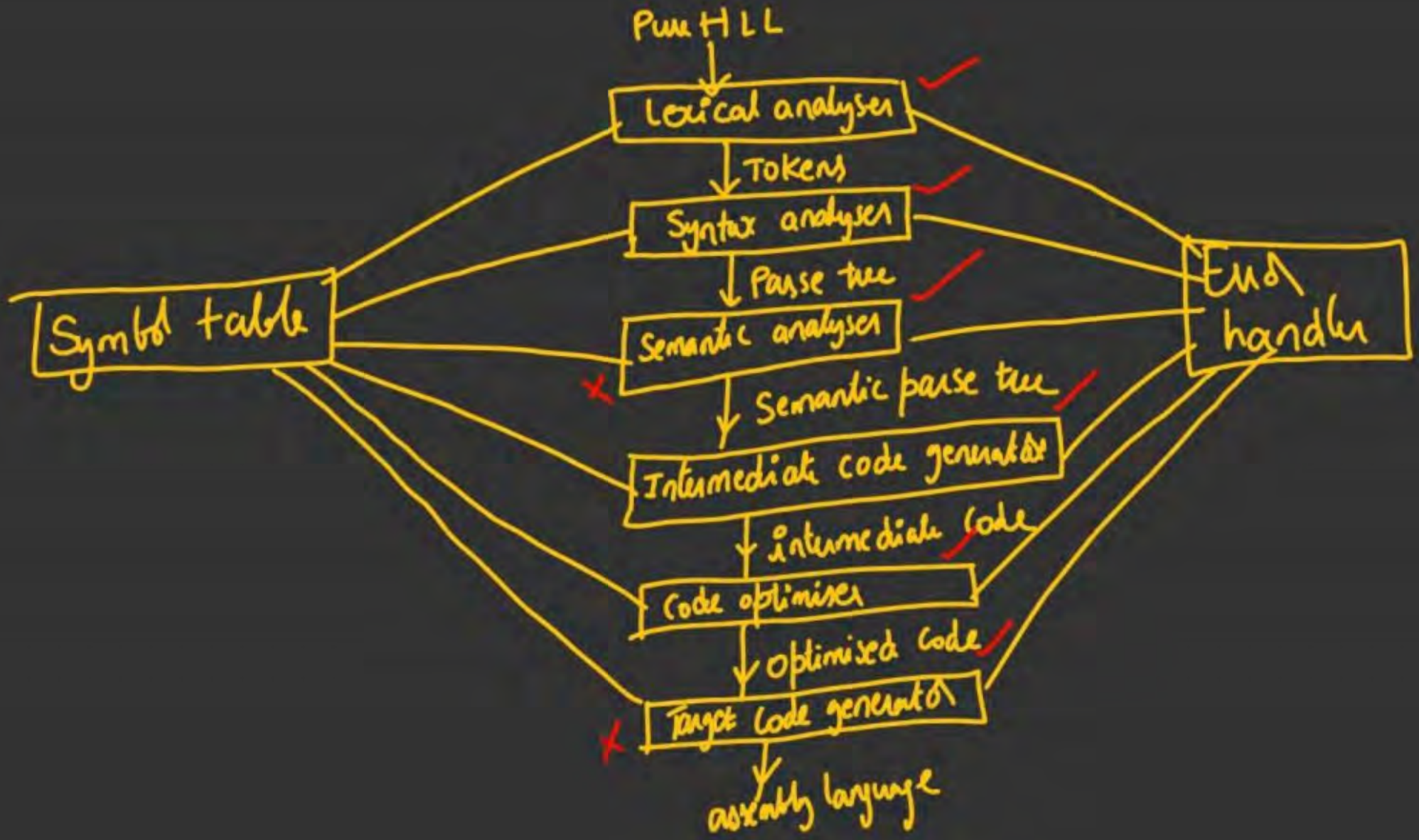
Linker :-

Linker may take libraries such as OS libraries & user defined libraries & language libraries

loader :-

This is a module which loads an executable file into memory, initializes the registers, heaps, data etc and starts the execution of the program

Addrss relocation is also done by loader


Pure HLL
Lexical analyser
Tokens
Syntax analyser
Parse tree
Semantic analyser
Semantic parse tree
Intermediate code generator
Intermediate code
Code optimiser
Optimised code
Target code generator
assembly language
Symbol table
Error handler

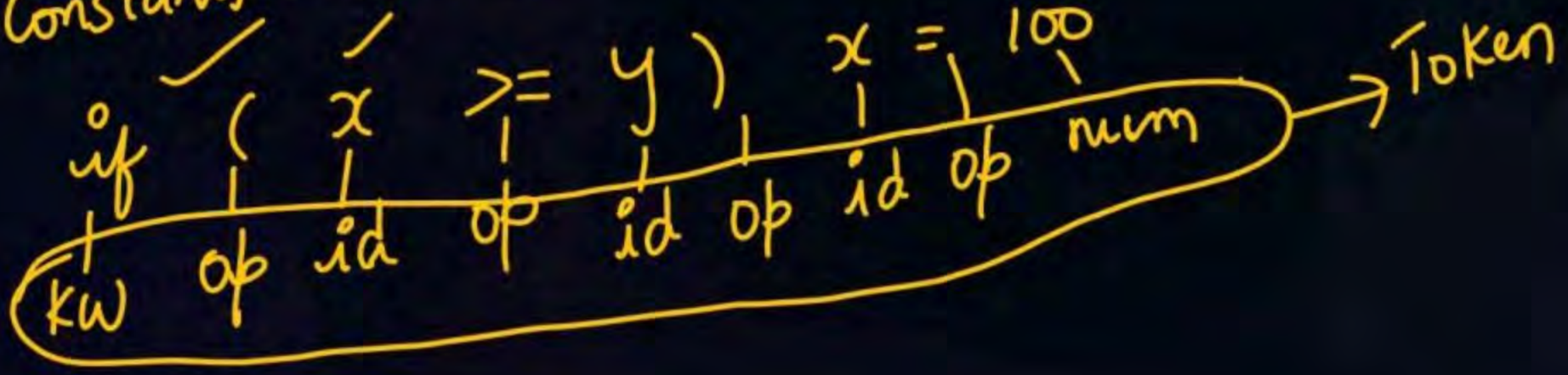
Design phases of Compiler :-

The lexical analyzer :-

Lexical analyzer is also called Scanner
The Scanner groups the i/p stream of characters into a stream of tokens and constructs the Symbol table, which will be used later

The tokens in any high level language can include

→ Keywords
→ identifiers
→ operators
→ Constants : numeric & character & strings



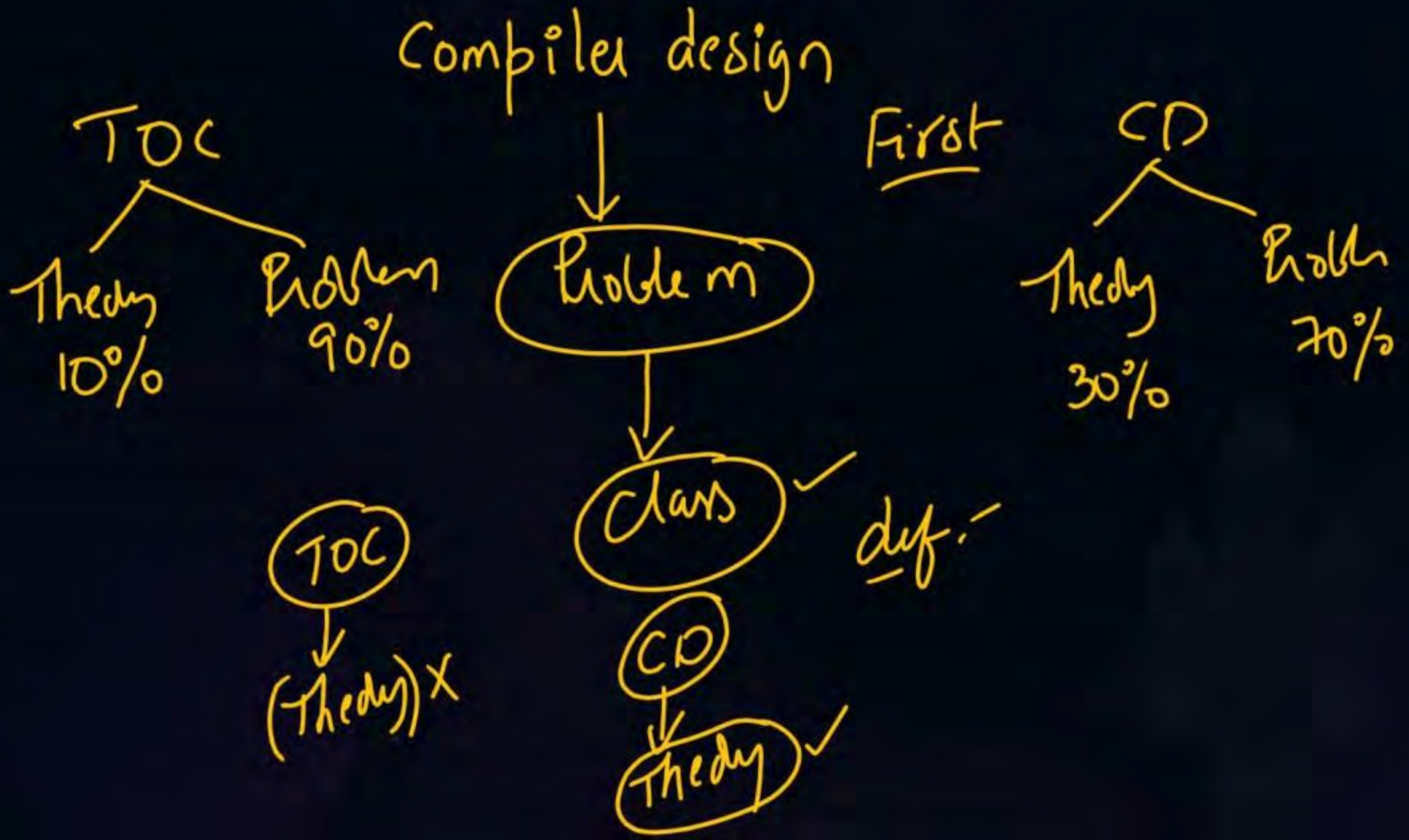
The i/p to lexical analyser is a character stream and o/p is stream of tokens.

Regular expressions are used to define the rules for recognising the tokens

lexical analyzer is implemented as a Finite automata.

Automated Scanner generation tools are lex and flex

RE → NFA → DFA


Compiler design
First
TOC
Theory 10%
Problem 90%
Problem
CD
Theory 30%
Problem 70%
TOC
(Theory) X
Class
def.
CD
Theory

15-days (2pm-5pm)

$15 \times 3 = 45$ hours

C - compiler

3 months

5 glade

Syntax analyser :-

Syntax analyzer checks the Syntax of a given Statement. The Syntax analyzer is also called parser The parser groups stream of tokens into a hierarchial structure called parse tree

id + id + id





Unstoppable Indians: Stories to Ignite Student Motivation

Verghese Kurien – The Milkman of India Verghese Kurien (1921–2012) led Operation Flood (1970–1996), creating the world's largest dairy development program. As founder of the National Dairy Development Board, he devised a three tier cooperative model—village societies, district unions, and a national milk grid—linking 15 million rural farmers to urban markets. He introduced modern pasteurization, chilling centers, and transparent pricing. Milk production soared from 20 to over 50 million tonnes annually, boosting rural incomes and nutrition. Kurien's blend of engineering rigor and social vision transformed India into the global leader in milk production.

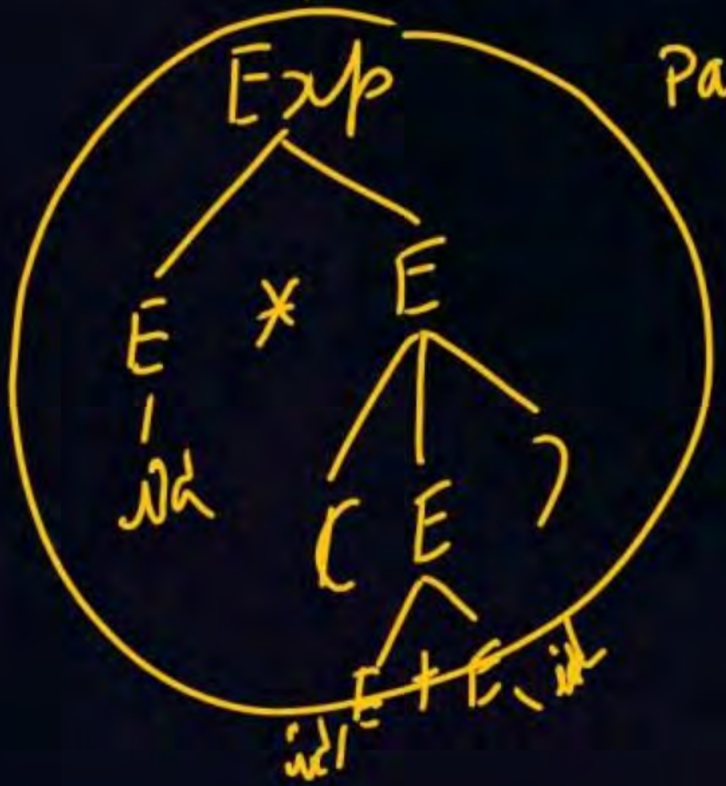
In order to check Syntax, first the Standard Syntax of all the Statements is to be known. Generally the Syntax Rules for any HLL are given by CFG.



Expr → Expr + Expr
/ Expr * Expr
/ (Expr)
/ num

CFG

6 * (8 + 9)



Parse

Semantic analyzer:-

Checks are performed to ensure that parts of a program fit together meaningfully.

Ex: (A + B)

Parser looks at binary addition and checks whether two operands are present. The presence of two operands confirms that the syntax is right. However, parser doesn't recognise the operand type.

A + B

Even if 'A' is array and B is a function, parser Says that Syntax is right.

It is the Semantic analyzer that checks whether the two operands are type compatible or not.

Static type checking is the basic function of semantic analyzer. During parsing, information concerning variables is stored in symbol table. This information is used by Semantic analyser to perform checking

Intermediate code generator :-

This phase converts the hierarchial representation of source code, that is parse tree into intermediate representation (IR) of the program.

Intermediate language should have two properties

a) It should be simple and easy to generate

b) It should be easy to translate to the target language

Intermediate can be represented in different way.

1) As a Syntax tree
2) Directed acyclic graph
3) Post fix notation
4) 3 address code

Code optimizer :-

Compiler time code optimization involves static analysis of intermediate code to remove extra operation.

The main goal of code optimizer is to improve the efficiency. Hence it always tries to reduce the number of instructions. once the number of instructions is reduced, the code runs faster and occupies less space.

Constant folding

$I = 4 + j - 5 \Rightarrow I = j - 1$

Constant propagation

$I = 3$
$J = I + 2$

$\Rightarrow I = 3$
$J = 5$

Code motion & frequently reduction

```
for(i=0; i<=5000; i++)
    a = sinx/cosx * i;
```

loop invariant

```
t = sinx/cosx
for(i=0; i<=5000; i++)
    a = t * i
```

Common Sub Exp elimination ⟶ DAG

$A = 100 * (B+C)$

$D = 200 * 8 + (B+C)$

$E = A + (B+C)$

$T = B+C$

$A = 100 * T$

$B = 200 * 8 + T$

$E = A + T$



Unstoppable Indians: Stories to Ignite Student Motivation

Prafulla Chandra Ray – Father of India's Chemical Industry Dr. Prafulla Chandra Ray (1861–1944) founded Bengal Chemicals & Pharmaceuticals in 1892, India's first pharmaceutical company. After studying in Edinburgh, he returned to Calcutta to develop affordable, indigenous drug manufacturing using local raw materials. He pioneered formulations for compounds like calcium carbide and analgesics, drastically cutting import dependence. Ray also established advanced research facilities and mentored a generation of Indian chemists. His entrepreneurial spirit and scientific leadership laid the foundation for India's modern pharmaceutical and chemical sectors.

Strength reduction

$2 \times x \Rightarrow x + x$

$2 \times x \Rightarrow x \ll 1$

Target Code generator:-

The code generator's main function is to translate the intermediate representation of source code to the target machine code. The machine code may be executable binary code of assembly code.

To design a target code generator, the designer should have complete knowledge about the target machine such as

a) Data handling
b) machine instruction syntax
c) variable allocation
d) Program layout
e) Registers
f) Instruction set



Source text :

if (x <= 3) then y = 2 * x

else y = 5

Target code : assembly

10) cmp x 3

11) JG 15

12) mul 2 x Temp

13) mov Temp y

14) Jmp 16

15) mov 5 y

Symbol table manager and Error handler:-

In addition to passing the data from one phase to another, additional information aquired during a phase may be needed by later phase. The Symbol table manager is used to store the record for each identifier, procedure encountered in the source langage with relevant attributes

int x = 10

id	type	addr	token
x	int		id

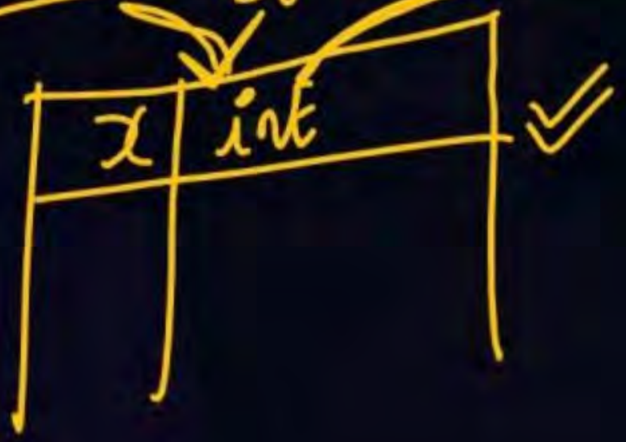
f(int a, int b, int c)

f	a	b	c	int	int	int	3 params

→ whenever any phase encounters new information, records are stored in symbol table.

→ whenever any phase requires any information, then records are retrieved from Symbol table

Ex: int x ; → syntax analysis → Semantic



Error handling:-

The error handler is used to report and recover from errors encountered in the source program. A good compiler should not stop on seeing the very first error. Each phase may encounter errors, so to deal with errors and to continue further, error handlers are required in each phase.

1) Logical error 2) Syntax error 3) Semantic error.

Unstoppable Indians: Stories to Ignite Student Motivation

Govindappa Venkataswamy – Aravind Eye Care Model Maker

(1918–2006) founded Aravind Eye Hospital in Madurai in 1976 to 'eradicate needless blindness.' He devised a high volume, low cost surgical model: standardized cataract procedures, factory style lens manufacturing, and a mixed patient paying system that subsidizes free operations. His franchise style expansion and rigorous training have restored sight to more than 5 million people and inspired sustainable healthcare solutions in dozens of countries. Professor Arogyaswami Paulraj (b. 1944) invented MIMO technology in 1992 at the Indian Institute of Science, Bangalore. He developed algorithms enabling multiple antennas at both transmitter and receiver ends to dramatically boost data rates and reliability. Through IIT Madras's technology transfer office, his patents paved the way for LTE and modern Wi Fi standards worldwide. Today's high speed mobile broadband networks rely on his work, and he has received numerous IEEE and governmental honors for revolutionizing wireless communications.

a) 1:15 to 4:15

b) 1:30 to 4:15

THANK - YOU